# A New Approach Toward Professional Responsibility

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**Abstract**

Nowadays, more than ever, technology is moving at a very high speed rate. With it, even professional figures and the way work is approached by big companies is evolving really fast. In this context, it's necessary to revisit the concept of professional responsibility. With this paper, I'm going to propose a new approach for assessing the responsibility someone has as a professional engineer. I'll show why this approach meets the paradigms and mentality of the current industrial world, while also trying to anticipate some possible counterarguments.

## 1 Introduction

The criterion used to attribute responsibility plays a significant role in an engineer's performance and in the results he produces. For this reason, it's important to update it to keep up with the evolution of professional standards. In previous works, several conditions have been proposed to evaluate responsibility, as wrong-doing, foreseeability or freedom of action [1]. First and foremost, these conditions allow us to decide whether an individual can be considered "responsible" for a certain situation. But they also allow us to understand to which extent he is responsible, to be able to estimate the seriousness or commendability of his actions. However, these are guidelines historically built on the basis of obsolete mindsets and principles of the engineering world. In this paper, I want to propose a new method to assess responsibility, under which an individual is always fully responsible for what he causally contributes to, regardless of the way he achieves a certain result or the intentions. As implication, the seriousness or commendability of his actions is only given by the consequences they cause. My goal is to show that it's always necessary to adopt this approach if we want the individual to work according to modern professional values.

In particular, I'll start by clarifying the meaning of responsibility and causal contribution. From there, I'll analyze the main modern work philosophy, showing why my approach better meets this philosophy than older approaches. Finally, I will present some possible objections that can be raised against my work, trying to answer them.

## 2 The Meaning of Responsibility

With "responsibility" I simply mean being held blameworthy for some unwanted circumstances, or even praiseworthy for the occurrence of a desired event [2] [3]. Different nuances of responsibility can be found in the literature [1]. A first, fundamental distinction is between *active responsibility* (responsibility before something has happened) and *passive responsibility* (responsibility after something has happened). Another possible classification is driven by the different contexts in which responsibility can be measured (professional, moral, or role-based context).
In this paper I will always implicitly refer to *professional passive* responsibility. "Professional" means that I'll focus on the responsibility someone has as a professional, in particular as a professional engineer. It's not in my intention to generalize my view to any other context. "Passive" means that I'll analyze how an individual can be held responsible for a certain result, only when that result has been produced. Nevertheless, I'm going to show how the way we judge passive responsibility impacts the way the individual approaches his job in the first place.

## 3 The Role of Causal Contribution

As widely shared by the philosophical community [1] [4], the basic condition for a person to be held responsible for a given event, is that he has causally contributed to the occurrence of that event. To understand the concept of "causal contribution", we can model it through this simple statement:

*A subject performs an action. The action causes a result.*

The "action" refers to any activity or decision a human being can take, such as drinking a coffee or driving a car. However, for our purposes, we are interested in professional actions related to a specific engineering context. For a software engineer, it could be the implementation of a new feature in a mobile application; for a mechanical engineer, the improvement of the efficiency of a certain hydraulic component. Please mind that even the decision not to act is usually considered an action.
The action performed by the subject generates a "result", that is, the new setting of reality that would not have existed if the subject had not performed the action in the first place, regardless of the intentions.
Whenever we have to attribute responsibility for a certain result, the "subject"

is the person to be investigated.

The essential question is whether, in investigating the subject's responsibility, it is sufficient to consider the result or should we also analyze the action. At the moment, we are used to looking at the action. There have been several attempts to propose conditions that touch both the way the action has been performed, and the motivations behind it (we can for instance mention wrong-doing, foreseeability or freedom of action, all conditions related to blameworthiness) [1]. They allow us to establish whether the subject is responsible for a certain result, and to which extent. The extent to which he is responsible can also be seen as the seriousness (in case of an unwanted result) or the praiseability (in the case of an expected result) of his actions. If we think about it, in a professional environment this plays a fundamental role, as it can be seen as the main root of promotions or firings.

But I firmly believe that this approach is no longer suitable nowadays. What I want to show from here on is how, in modern society, the only reasonable choice is to ignore the action to just focus on the result. In other terms, the subject is always fully, totally responsible for the result he produces. This means that an action is a serious or praiseworthy as the result it cause.
Of course, saying "in modern society" is very vague, and it would be difficult (if not impossible) to prove that my thesis holds for every professional field, company culture and place in the world. For this reason, I decided to proceed as follows: I am going to identify the most revolutionary professional culture, adopted by today's most successful companies. I'll then show that the proposed approach is the best choice to complement this culture. By ascertaining that such a culture represents most of the professional world not only today, but ever more in the future, the validity of my thesis will follow.

# 4 An Evolving Philosophy of Work

In the last decade, many big tech companies such as Google, Meta, Amazon or Tesla have based their success on a work philosophy built around moving fast and failing fast. It's the classic tech mindset: prioritize rapid iteration, embrace risk, and accept failure as a necessary part of innovation [5]. We can break down this philosophy by isolating the method it proposes, and the result it aspires to obtain.

***Method***
The method consists of three ingredients: *risk*, *failure tolerance* and *speed*.
One must constantly take risk and fail, attempt after attempt, without worrying about what the final result will be. Moreover, this has to be done by moving as quickly as possible, favoring a fast result over a perfect one.

***Objective***

By the experience and skills acquired through this process, in the long run the company will boast products of the highest quality, obtained much earlier than companies that focus on great results in the short term.

It's important to highlight that it's not the purpose of this paper to discuss whether this approach is good or bad. For instance, it may not be a good idea for mission-critical businesses, like pharmaceutical companies. The main point here is to assert its relevance and the fact that it seems to have proved successful for its adopters, reason why it appears to be rapidly expanding [6].

# 5 The Importance of Responsibility Attribution

The way I want to proceed now is by presenting a thought experiment that can be concretely replicated in many real life situations. In this thought experiment, we have two subjects, or employees, that are working on the same long-term project, no matter what it is. We can assume both subjects have the same skills, experience and education background, so that neither has an advantage over the other. We then imagine that there is no way in which the two can come into contact, for the entire duration of their work. This means they can't communicate, exchange opinions or motivations. They can for example work in different companies, or in completely separate sections of the same company, under a different management.
Both are supposed carry on the project by adopting this "move fast and fail fast" approach.
Once the projects have been completed, they are judged "responsible" to some extent for what they have produced (whether good or bad) according to a chosen criterion. For the entire duration of their work, both subjects are aware of the criterion that will be used. Two things will happen:

- They will be rewarded in proportion to what they have produced good, and to the extent they are responsible for it;
- They will be penalized in proportion to what they have produced bad, and to the extent they are responsible for it;

Their goal is to maximize the reward and minimize the penalty.
The only difference between the two subjects is that, for each of them, we use a different criterion to attribute responsibility. We evaluate the responsibility of the first subject by analyzing the course of actions that led to the final outcome. In the second case, instead, the subject is always fully responsible of the outcome.

The aim is to highlight how this has an impact on the way each subject approaches his task, and in which case the performances are maximized.

About the first subject, we can for instance verify that he followed the standard procedure without violating any norm, or if the intentions matched the final result. This would mean potentially justify him for the occurrence of a

certain bad situation, or, conversely, (partially) deprive him of the recognition for a success, if it was achieved in a way considered "inappropriate".
In response, the subject will behave accordingly, by also placing his attention on the actions he carries out to complete the project. What consequences do we observe on his "move fast" approach? Let's break it down:

***Method***
The subject will try as much as possible to follow a course of action that is in line with a certain standard. He will also make decisions that can be sufficiently motivated, refraining from hunches and intuitions. As a consequence, the risk factor is almost completely absent.
Another remarkable aspect is the foreseeability: to be able to justify his choices a posteriori, the subject will not take any action without being aware of the consequences. This is in contrast to the "fail fast" of "doing something without worrying about the outcome" mentioned earlier, upon which the failure tolerance factor is based.

***Objective***
Following a course of action that doesn't deviate too much from the norm, the subject's approach to his job won't undergo significant mutations. As a consequence, the quality of what he produces is not going to improve much over time. In other words, he'll not reach the objective of retaining the highest quality product in the long run.

In the second case, the result of the subject's causal contribution is all we need to know. Inevitably, the subject's focus will entirely be on the final value of its realization. Even in this case, we evaluate the effects on the "move fast" method and objective:

***Method***
As a first effect that we can imagine, the subject will exploit his skills or experience by using his intuition to achieve the desired scope, instead of relying on the "correct way of acting". This has a significant impact on the execution speed.
We can then reason on the risk factor. We all know, from life experience, that some of the most ambitious goals have risk as the price to pay. If the result is all that matters, we can expect the subject to have the courage to break through this barrier, by embracing risk without much hesitation.
In turn, risk leads to failure tolerance, as by definition risk carries the possibility of failure.

***Objective***
Over time and during each attempt, the subject will adjust his working strategy with the sole purpose of improving the outcome. The effect is a faster progress and better results in the long run.

In summary, we discerned that looking at the action performed by the subject leads to a reduction in the *risk* and *failure tolerance* factors of this method, and prevents the achievement of the objective. On the contrary, evaluating only the outcome incentivizes the *speed*, *risk* and *failure tolerance* factors, prompting the achievement of the long-term objective.

We can now draw the conclusions of our thought experiment: with equal preparation, skills and experience between the two subjects, by simply shifting the evaluation of responsibility to the new approach, the second subject will better meet the requirements of performance and job management.

To recap, my argument can be formulated as follows: I started by proposing a new method for assessing responsibility, based on the mere observation of the result of someone's causal contribution. To prove its necessity in modern times, it was necessary to focus on the prevailing tech philosophy of "move fast and fail fast". The analysis showed that this method is the best option to assist such a philosophy. Furthermore, the novelty and success of this culture suggest its expansion, and this would extend the validity of the proposed approach to a more general scenario.
But I understand that for someone this may not be enough. This is indeed a sensitive topic, and one can raise different counter-arguments that can question my thesis. For this reason, in the next section I will try to anticipate and answer some of the possible objections.

# 6 Contrary Views

I will present the potential contrary views in the following fashion: the first two objections refer to possible reasons why it might be useful to evaluate the actions performed by the subject. The following objections, instead, go more specifically to the argument in support of my thesis. In particular, the third one argues that my approach to responsibility is not really useful under "move fast and fail fast" conditions, while the fourth under any other condition.

## 6.1 The Threat of Unethical Behavior

It could be claimed that, although it promotes speed and quality, using the result as the only judgment metric is unethical, as it justifies immoral and bad actions performed for the sole purpose of obtaining the desired result.

This is an interesting objection, but it needs to be further analyzed. For instance, one may argue if the action can be really defined as "immoral" if useful to obtain a good result. Can a doctor's disobedience to the hospital rules be considered immoral if it saves a patient's life? We are therefore led to make a distinction between what is good for a company and what is good for the

society or the welfare of the people around us. In this framework, causing physical or moral harm to another person for the purpose of bringing more money into the company coffers can be safely consider "immoral".
But if we do this distinction, we must do it also for the results of the subject's causal contribution. In other terms, the subject causally contributes to bringing more money into the company coffers, but he also causally contributes to causing harm to the person. The damage to the welfare of the society is no longer an action, but becomes another result of the action, for which the subject must be held responsible in the same way.
We can of course say that some modern companies ignore this kind of results, but this is a debate that departs from the topic of our discussion, and it's completely independent from the way we attribute responsibility.

## 6.2 Freedom of Action

This argument states that the subject cannot be considered responsible whenever he acts under coercion, or against his will. Indeed, we've seen that a subject is only responsible for what he causally contributes to. But if the causal contribution has been forced, it would be illogical to still consider him "responsible".

I am not of the opinion that an individual is less responsible for what he causes whenever he doesn't act freely. But in any case, this objection has actually little to do with the thesis of this paper. Here's why: dealing exclusively with professional responsibility, it mainly refers to the case where the employee is threatened with being fired or not promoted. However, even the supporters of this objection are ready to claim that the coercion must be stricter than a simple salary cut [1]. Even if we wanted to consider it as a valid coercion, at that point we should balance professional responsibility with the moral (and role) responsibility that the employee has in providing a salary for himself and/or his family. Again, we would go off topic, and this would require a more extensive study.
For this reason, I am not going to discuss this objection further.

## 6.3 The "Inconsistency" Objection

This argument is intended for all those who argue that the thought experiment presented above is wrong, and so the new proposed approach to responsibility is not the best choice to incentivize a "move fast and fail fast" philosophy.
The reason behind this theory may be the following: by evaluating responsibility only on the basis of the result, the subject will be more afraid of making mistakes, even when he is aware of his own potential. He'll thus avoid taking risks and moving quickly, negatively affecting the method and, in consequence, the long-term objective.

This objection can be easily answered by clarifying, once again, the meaning of this philosophy. Like any other corporate culture, a "move fast and fail fast" culture aims to achieve high quality results. However, the key difference is that it only requires them over the long term. In the short term, whenever the subject approaches a new venture, failure is not only accepted, but also encouraged. Actually, this attitude is really meant to undermine the fear of making mistakes. Of course, this does not mean that the subject will be less responsible for his own results in the short term. It just means that he won't be punished for them. As a consequence, he will be driven to move freely and boldly in pursuit of his goals.
This is the reason why, in the proposed thought experiment, the subject is rewarded or penalized only upon completion of his project.

### 6.4 The "Rest of The World" Objection

At this point, many could be convinced of the validity of my thesis in a "move fast and fail fast" context. But they could argue that, as much as this culture may expand, it's currently not adopted everywhere. Even in the future, it's reasonable to think that there will always be areas where it'll not be introduced. For this reason, my thesis cannot be generalized, as it would lead to undesirable results in the rest of the world.

I want to start answering by reiterating why I decided to focus on a single business culture. The aim was *not* to "find a business culture that needs a new approach to assessing responsibility", but instead: "in order to prove the need for a new approach to assessing responsibility, consider the most influential business culture to show that, in such an environment, this is true.". In other words, it was only a practical matter, as it would have been impossible to focus on every possible engineering context. That said, we can realize that the way this objection is worded is incorrect. It assumes that, wherever this philosophy is not adopted, my thesis is incorrect. But this is not necessarily true. It just means that its effectiveness has not yet been proven or disproven. Actually, there are no other ways in which I can reply, except by saying that, in case the study will be extended to other industrial realities, it is plausible that my thesis will be further confirmed.

## 7 Conclusion

In this paper, I presented a new method of assessing professional responsibility. I showed how it can benefit the current paradigms of the corporate world, anticipating possible contrary views. From here, we also highlighted that the work cannot be considered concluded, due to the different professional realities that surround us. The consequent logical research direction consists in extending the analysis of the proposed approach to other modern realities.

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